



A novel hybrid model with two-layer multivariate decomposition for crude oil price forecasting

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ABSTRACT

Crude oil plays an important role in economic development and political stability, and many scholars have been committed to forecasting its price. However, its influencing factors are complex and diverse, and previous studies have rarely focused on the second multivariate decomposition. Therefore, this study introduces financial market factors and crude oil news as forecasters, and proposes a novel hybrid model with two-layer multivariate decomposition. To verify the performance of the proposed model, an empirical study is performed on weekly West Texas Intermediate (WTI) oil spot price. The results suggest that the second multivariate decomposition for the high-frequency subcomponent can significantly improve the forecasting accuracy, and the forecasting performance of the proposed model outperforms all the benchmark models.

1. Introduction

Crude oil plays an indispensable role in people's daily life and industrial economic development, and also regarded as an important strategic resource worldwide [1]. Given that crude oil is so important, the precise forecasting of its price can help relevant participants to make rational decisions. However, the complexity of crude oil supply and demand structure and the diversity of its influencing factors give rise to the difficulty of crude oil price forecasting [2].

Therefore, for the difficulty above, many scholars have carried out extensive research, and the majority of them focused on the research of hybrid forecasting model with multiscale decomposition. For example, Jammazi and Aloui [3] established a hybrid forecasting model with wavelet transform (WT). However, its mother wavelet function and the number of decomposition layers are difficult to determine. To overcome the shortage of WT, empirical mode decomposition (EMD) and its improved form were developed and widely used [4–6]. Although the models with EMD family have achieved some good results, the end effect of EMD family may affect the decomposition effect. By contrast, variational mode decomposition (VMD) is favored by the academic community, and has been proven to be better than EMD [7,8]. Meantime, as the

crude oil price is affected by many factors, scholars have also studied the hybrid forecasting models with multivariate decomposition [9–11]. At present, the commonly used multivariable decomposition methods include multivariate EMD (MEMD) and multivariate VMD (MVMD) [12, 13]. However, most scholars are interested in the model with MEMD. Moreover, few research focused on second multivariate decomposition. Therefore, this study adopts MVMD to establish a forecasting model with two-layer decomposition-ensemble. Additionally, as forecasting performance of the decomposition-ensemble model is often affected by irregular events [14], crude oil news reflecting irregular events is regarded as a predictor.

In summary, this study utilizes the advantages of the methods and devotes to overcome the weaknesses of previous studies, and proposes a new hybrid model, which incorporates two-layer multivariate decomposition and news text. Based on the model above, this study mainly aimed to (1) verify whether the two-layer multivariate decomposition can improve the forecasting performance of crude oil prices, (2) verify whether the forecasting effect of the model with MVMD is better than that of the model with MEMD, and (3) verify the superiority of the proposed model.

The main contributions and innovations of this study include the

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following aspects.

- (1) A novel hybrid model with two-layer multivariate decomposition is proposed. Specifically, this study may be the first to apply the two-layer multivariate decomposition strategy for crude oil price forecasting. Additionally, the crude oil news is incorporated into a new multivariable two-layer decomposition-ensemble framework.
- (2) The decomposition effects of MEMD and MVMD were compared. The results suggest that although the MVMD-based model is superior to the MEMD-based model, the advantage is not very noticeable, which may be related to the number of sub-modes.
- (3) Built upon different results of Granger causality test, different methods are adopted to determine the lag order. Specifically, if the subcomponents have corresponding forecasters, the lag criterion of the vector autoregressive (VAR) model is utilized to determine the lag order. Otherwise, phase space reconstruction (PSR) or partial autocorrelation function (PACF) is adopted to determine the lag order according to the chaotic characteristic of the subcomponent.

The remainder of this study is organized as follows: Section 2 is a review for the relevant literatures. Section 3 describes the main methods used. Section 4 presents the empirical results and analysis, and Section 5 displays the study conclusions and limitations.

2. Literature review

In recent years, the researchers have mainly improved the forecasting accuracy of crude oil prices by perfecting predictive approaches and predictors. For the forecasting methods, they can be roughly divided into three categories. The first includes the econometric models represented by autoregressive integral moving average (ARIMA) and generalized autoregressive conditional heteroscedasticity (GARCH) group [15,16], and the artificial intelligence (AI) models represented by support vector machine (SVM) and artificial neural networks (ANNs) [17–19]. As is well known, although the econometric models are simple in structure and easy to analyze, they are built on restrictive distribution assumption [20]. By contrast, AI models have no strict assumption and are better suited to handle complex data. However, it is difficult to achieve good forecasting results by using only one forecasting technique for all data [21].

The second is combination model, which combines the forecasting values of different models via weight to achieve the final forecasting result; this idea was first proposed by Bates and Granger [22]. Yu et al. [23] adopted least squares support vector regression (LSSVR) with different parameters to forecast crude oil prices. It turned out that the LSSVR ensemble learning model had the best forecasting effect. Azevedo and Campos [24] combined ARIMA and exponential smoothing models to forecast crude oil prices, and the results indicated that the combination model achieved preferable forecasting results. Zhang et al. [25] proposed an iterated combination model, and the results proved that the proposed model obtained better forecasting results. Wang et al. [26] first used three different individual models, including multiple linear regression (MLR), SVR, and ANN, to generate forecasting value; further, these values were combined to get the final forecasting result. Although this type of model combines the advantages of different models and improves forecasting accuracy to a certain extent, it is incapable of effectively capturing the deep nonlinear characteristics concealed in crude oil prices.

The third is the hybrid model with decomposition-ensemble idea. This type of model can clearly present the relationship between frequency domain and time domain through time-frequency analysis, and has evident advantages in mining the inherent features of complex data [27]. According to the considerable existing literatures, hybrid models can be divided into two cases: univariate decomposition-ensemble [5,

28–31] and multivariate decomposition-ensemble forecasting models [32]. For example, Zhu et al. [5] developed a new adaptive hybrid model with univariate decomposition-ensemble. Tang et al. [32] synthesized the internet search data into multiple indexes, and proposed a hybrid model with multivariate decomposition-ensemble. Overall, this type of model not only combines the advantages of different models but can also capture the nonlinear characteristics of crude oil prices effectively.

However, looking at the previous studies on hybrid forecasting model, we find that the majority of them used MEMD to build decomposition-ensemble forecasting model, and few studies have attempted two-layer multivariate decomposition. Therefore, considering the advantages of forecasting methods and the shortcomings of existing studies, this study proposes a novel two-layer decomposition-ensemble model with MVMD to fill a possible gap, which may become a new application prospect in the field of oil price forecasting, and is expected to improve the forecasting precision of crude oil prices.

From the perspective of predictors, according to the existing research, the predictors can be classified into three groups: crude oil market, financial market, and unpredictable factors that upset the equilibrium of the market [33].

Crude oil and finance market: The complex supply (e.g., production, inventory, and export) and demand (e.g., import and global GDP) structure of the crude oil market is an important factor affecting the volatility of crude oil prices. Many studies have verified their role in crude oil price forecasting [28,34–36]. For example, Gogolin et al. [35] verified the existing co-integration relationship between crude oil prices and the economy, in which global GDP was regarded as a proxy variable of the economy. Meantime, there is also a relationship between crude oil and financial markets (e.g., exchange rate, stock, and futures), which has been verified by many existing studies [37,38].

The abovementioned two groups can be summarized as structured data. However, the existence of an active derivatives market is an important feature of crude oil market [39]. Therefore, there is another group of unstructured influencing factors: investor attention and investor sentiment. For the former, prior research generally regarded Internet search data represented by Google Trends and Baidu Index as a proxy of investor attention [40]. For the latter, text data represented by official news and user-generated comments were quantified as sentiment scores to stand for investor sentiment [41]. Moreover, many researches have verified that these two types of unstructured data contribute to the analysis and forecasting of crude oil prices [14,42]. As it is, the conclusion above has also been verified in other fields [43–46].

For the influencing factors, it is indisputable that the forecasting performance of crude oil prices is affected by the first two influencing factors. However, the decomposition-ensemble framework is likely to suffer from poor predictive performance when unexpected events (e.g., geopolitics, conflicts, and wars) have an impact on the crude oil market [14]. Therefore, to improve the forecasting accuracy of crude oil prices, online oil news reflecting unexpected events is incorporated into the forecasting model.

3. Methodology

This section introduces main methods utilized in this study and the system design of the proposed model. As kernel principal component analysis (KPCA) [47], forecasting techniques (MLR, backpropagation neural network (BPNN)), and some statistical test methods (e.g., VAR, PACF, and Granger causality test) are the most basic method, they will not be further elaborated here.

3.1. Multivariate variational mode decomposition

MVMD, proposed by Rehman and Aftab [13], is an extension of VMD. Its coordinate system is changed from a two-dimensional to a spherical coordinate system. For the original forecasting target and predictors $X(t) = [x_1, x_2, \dots, x_C]$, it is assumed that they are decomposed

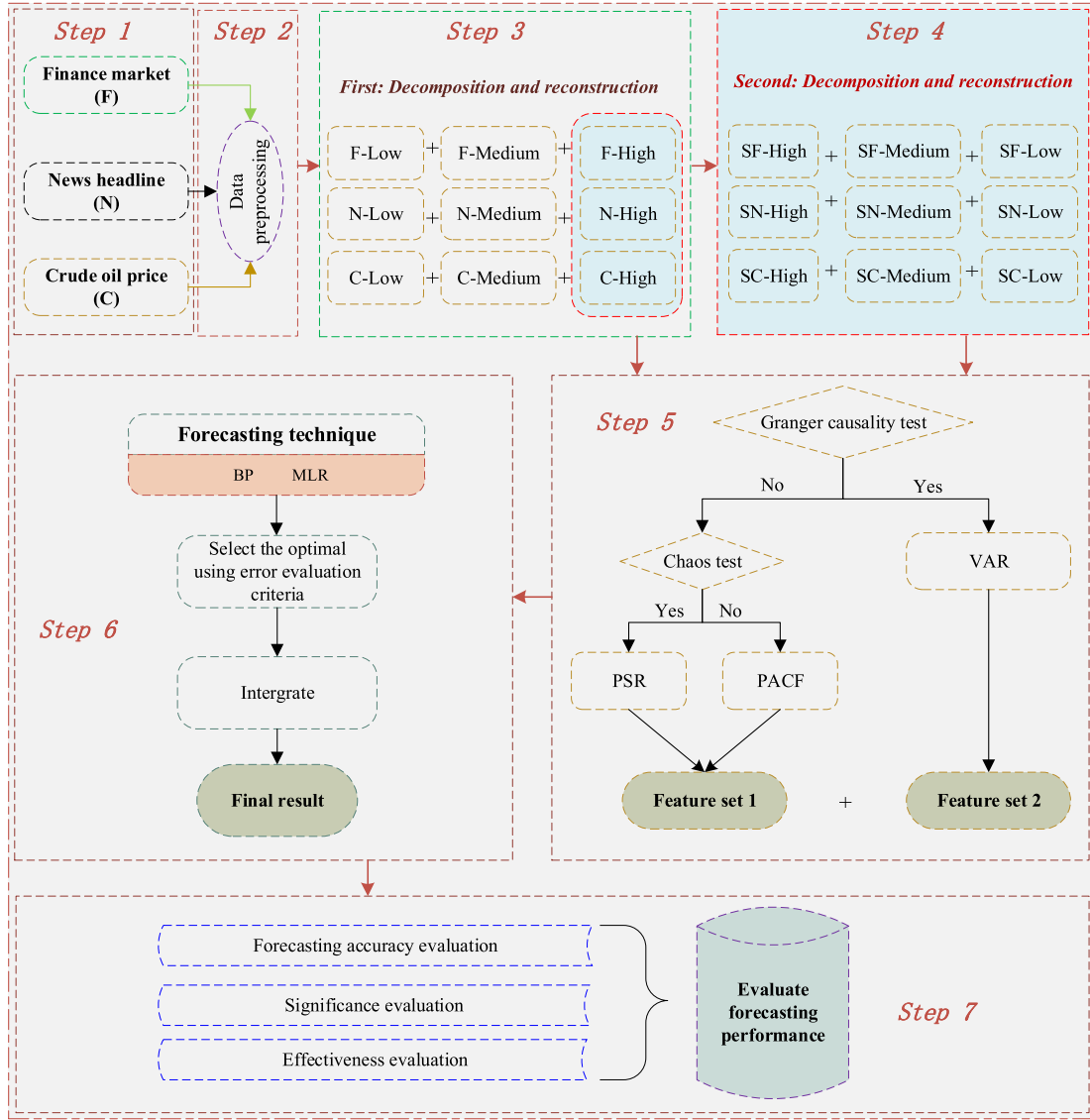


Fig. 1. Flow chart of the proposed model.

into K number of multivariate intrinsic principal modes $u_k(t) = [u_1, u_2, \dots, u_c]$, where $k = 1, 2, \dots, K$. Two criteria are followed when selecting modes: one is that the sum of bandwidths of all modes is minimum, and the other is that the sum of all the modes is equal to the input signal. Therefore, the constrained variational model problem is expressed as follows:

$$\begin{cases} \min_{\{u_{k,c}\}, \{\omega_k\}} \left\{ \sum_{k=1}^K \sum_{c=1}^C \left\| \partial_t [u_{k,c}^{k,c} e^{-j\omega_k t}] \right\|_2^2 \right\} \\ \text{s.t. } \sum_{k=1}^K u_{k,c}(t) = x_c(t), c=1, 2, \dots, C \end{cases} \quad (1)$$

where $u_{k,c}$ and ω_k represent the k th mode and center frequency of the c th time series, respectively. ∂_t refers to partial derivative operation regarding t . $u_{k,c}^{k,c}$ is the analytic modulated signal of $u_{k,c}$, which is obtained via the Hilbert transform, and $j^2 = -1$.

Furthermore, the constrained optimization problem above is transformed into an unconstrained optimization problem. Thus, formula (1) can be rewritten as formula (2), where λ and α refer to the Lagrange multiplier and the equilibrium parameter, respectively.

$$\begin{aligned} \mathcal{L}(\{u_{k,c}(t)\}, \{w_k(t)\}, \lambda_c) = & \alpha \sum_{k=1}^K \\ & \times \sum_{c=1}^C \left\| \partial_t [u_{k,c}^{k,c} e^{-j\omega_k t}] \right\|_2^2 + \sum_{c=1}^C \left\| x_c(t) - \sum_{k=1}^K u_{k,c}(t) \right\|_2^2 + \sum_{c=1}^C \langle \lambda_c(t), x_c(t) \\ & - \sum_{k=1}^K u_{k,c}(t) \rangle \end{aligned} \quad (2)$$

Finally, the alternate direction method of multipliers algorithm is adopted to solve the problem in formula (2). Specifically, the unconstrained optimization problem is transformed into three sub-optimization problems with regard to $u_{k,c}(t)$, $w_k(t)$, and λ_c , and their updated formulas are expressed as follows:

$$\hat{u}_{k,c}^{n+1}(\omega) = \frac{\hat{x}_c(\omega) - \sum_{i \neq k} \hat{u}_{i,c}^{n+1}(\omega) + \frac{\hat{\lambda}_c(\omega)}{2}}{1 + 2\alpha(\omega - \omega_k)^2} \quad (3)$$

$$\omega_k^{n+1} = \frac{\sum_c \int_0^\infty \omega |\hat{u}_{k,c}^{n+1}(\omega)|^2 d\omega}{\sum_c \int_0^\infty |\hat{u}_{k,c}^{n+1}(\omega)|^2 d\omega} \quad (4)$$

$$\hat{\lambda}_c^{n+1}(\omega) = \hat{\lambda}_c^n(\omega) + \tau \left(\hat{x}_c(\omega) - \sum_k \hat{u}_{k,c}^{n+1}(\omega) \right) \quad (5)$$

where τ is an update parameter.

3.2. Text mining for news headlines

In the previous research, many scholars have studied the quantification of text data [14,41,42,48,49]. For example, Li et al. [14] quantified crude news texts into emotional polarity scores through a NLP library (Textblob). Al et al. [48] used natural language processing (NLP) technology to classify the tweets as subjective opinions or objective facts. In this study, referring to the literatures [14,41,42], crude oil news headlines with keyword are obtained, and each news headline is converted to emotional polarity scores through two Python libraries, *Textblob* and *Vader*, which are two classic sentiment analysis libraries in NLP. Specifically, first, these two libraries are used to calculate sentiment polarity scores separately, then the sentiment scores obtained are summed and averaged to get the final news sentiment scores. The final sentiment scores are within the range $[-1.0, 1.0]$, where less than zero means negative news, greater than zero means positive news, and zero means neutral news.

3.3. Lyapunov exponent

Sensitivity to initial conditions is a basic feature of chaotic motion, and the size of the Lyapunov exponent can reflect the sensitivity of the system to initial conditions [50]. Any system containing at least one positive Lyapunov exponent is defined as chaotic [51]. In generally, only the maximum Lyapunov exponent (MLYE) is calculated, and if MLYE is greater than zero, the system has chaotic characteristics. Therefore, the Wolf method is adopted to calculate MLYE, and the detailed description for Lyapunov exponent and Wolf method can be found the literature [52].

3.4. The proposed model

The model proposed in this paper consists of seven steps, as shown in Fig. 1, and the specific steps are described as follows:

Step 1. Collection data

The data sets collected in this study include crude oil spot prices, financial market, and crude oil news text data. Among them, the first is the forecast target, and the last two are exogenous predictors of the forecast target.

Step 2. Data preprocessing

To prevent the possible multi-collinearity problem and obtain investor sentiment, this step mainly involves the fusion of financial market data using KPCA and the quantification of news text using text mining means. As a result, the forecasters are obtained.

Step 3. First multivariate decomposition and reconstruction

First, crude oil price series, and all the forecasters obtained in the Step 2 are simultaneously decomposed into several sub-modes using MVMD. Then the sub-modes are reconstructed into high-, medium-, and low-frequency subcomponents via sample entropy (SE) and hierarchical cluster analysis (HCA).

Step 4. Second multivariate decomposition and reconstruction

The same decomposition and reconstruction process as in Step 3 is performed for the high-frequency subcomponents. Eventually, the high-frequency subcomponents in step 3 are decomposed and reconstructed into new three groups of subcomponents. It is worth noting that "Second" is denoted as "S" in the step 4 of Fig. 1. For example, "SF-High" represents the financial high-frequency subcomponent after the second decomposition and reconstruction.

Step 5. Feature selection

This step screens effective predictors. First, we test the Granger causality relationship between the subcomponents of forecasting target and the subcomponents of the corresponding predictors. Built upon the tests above, two possible results are obtained: one is that there is at least one Granger causality relationship, namely the subcomponents of crude oil price have at least one predictor with predictive power; the other is that there is no Granger causality relationship at all, namely the subcomponents of crude oil price have no predictors with predictive power. For the first result, the lag criterion of the VAR model are used to determine the lag order. For the second result, the MLYE is utilized to judge the chaotic characteristics of the subcomponents; if they are chaotic, PSR is adopted to determine their lag order; otherwise, the PACF is used to determine. Finally, two feature sets (Feature sets 1 and 2) are obtained.

Step 6. Crude oil price forecasting

As the subcomponents of crude oil prices become more predictable after two-layer decomposition and reconstruction, two classical forecasting techniques with relatively simple structure, MLR and BPNN, are used to forecast them separately. Further, the optimal individual forecasting results are selected using error evaluation criterions. Finally, the values selected are integrated to obtain the final forecasting result.

Step 7. Performance evaluation

This step evaluates the performance of the proposed model from three perspectives: forecasting accuracy, significance, and effectiveness. First, the forecasting error evaluation indexes of level and direction are adopted to evaluate the forecasting accuracy of the proposed model. Further, the Diebold-Mariano (DM) test [53] is employed to test whether the proposed model is significantly better than the benchmark models from the perspective of statistics. Finally, the improvement ration (IR) for forecasting accuracy is calculated to evaluate effectiveness, which is widely used by other studies [41,54].

4. Empirical study

4.1. Empirical design

(1) Data description

The WTI spot price (WTI price) is the forecasting target, with the data achieved from well-known oil website "eia.gov". These data cover September 5, 2014, to November 26, 2021. The corresponding weekly financial variables are collected from the Federal Reserve Economic Data website. Specifically, they include the Dow Jones Industrial Average Index (DJIA), S&P 500 Index (S&P500), Nasdaq Composite Index (NASDAQC), Broad US Dollar Index (BUSDX), US Dollars to Euro Spot Exchange Rate (USRUE), and crude oil futures closing price. Obviously, these variables represent the stock, exchange rate, and crude oil futures market. Many studies have shown that they have an impact on the crude oil market [39], which is why these variables are selected in this study. The corresponding crude oil news are retrieved from the popular energy portals "oilprice.com" and "investing.com". As the crude oil news headlines are easier to retrieve and short headlines contain fewer repetitions and irrelevant words than the whole news document [14], this study utilizes news headline instead of the whole news document. Eventually, 4761 oil news headlines with the keywords "Crude oil" and "oil price" are obtained, in which 1803 headlines are based on the keyword "Crude oil", and 2958 headlines are based on the keyword "oil price".

Additionally, the abovementioned datasets are divided into training (the first 80% of the datasets) and testing sets (the last 20%), and used to train and test the models, respectively.

(2) Parameter and benchmark model setting

Table 1

Parameter values of the main algorithms.

Algorithm	Parameter	Value	Algorithm	Parameter	Value
MVMD	Bandwidth constraint	2000	BPNN	Epochs	200
	Number of modes	8		Batch size	10
SE	Embedding dimension	2	Learning rate	Learning rate	0.005
	Similarity tolerance	0.2SD		Hidden neurons	5

Since MEMD can adaptively determine the number of subcomponent, this study attempts to determine the number of MVMD subcomponent using MEMD. The parameters of the other main algorithms are determined by trial and error, and the specific settings are shown in Table 1.

In the introduction, three problems that need to be solved in this study are put forward, namely the superiority of MVMD method and the proposed model, and the effectiveness of MVMD two-layer decomposition strategy need to be verified. To solve the three problems above, this subsection sets up benchmark models from the following three aspects: different decomposition times, decomposition methods, and forecasting techniques. Specifically, the benchmark models with the first decomposition, MEMD, EMD, and VMD are set under the premise of using the BPNN, MLR, and hybrid forecasting techniques.

(3) Performance evaluation criteria

The performance evaluation criterions have been briefly introduced in Step 7 of Section 3.4. Therefore, this subsection describes them in detail.

The root mean squared error (RMSE), mean absolute error (MAE), mean absolute percentage error (MAPE), and direction accuracy (DA) [55] are utilized to evaluate the models' level and direction forecasting accuracy. Meanwhile, Pesaran-Timmermann (PT) test [56] is adopted to test the direction forecasting ability of the model from the statistical perspective. The methods above are expressed as Eqs. (6)–(10). Additionally, DM test and IR are used to analyze the significant and effectiveness of the models, in which the statistic of DM test is expressed as Eq. (11), and IR is expressed as Eqs. (12)–(14).

$$RMSE = \sqrt{\frac{1}{N} \sum_{t=1}^N (y_t - \hat{y}_t)^2} \quad (6)$$

$$MAPE = \frac{1}{N} \sum_{t=1}^N \left| \frac{y_t - \hat{y}_t}{y_t} \right| \times 100 \quad (7)$$

$$MAE = \frac{1}{N} \sum_{t=1}^N |y_t - \hat{y}_t| \quad (8)$$

$$DA = \frac{1}{N} \sum_{t=2}^N d_t \quad (9)$$

where N is the total number of test samples. y_t refers to the actual value, and $\hat{y}_t$ refers to the forecasted value at time t . If $(y_t - y_{t-1})(\hat{y}_t - \hat{y}_{t-1}) > 0$, then $d_t = 1$; if $(y_t - y_{t-1})(\hat{y}_t - \hat{y}_{t-1}) < 0$, then $d_t = 0$.

Table 2

Descriptive statistics of news sentiment scores.

Keywords	Before Processing	After Processing	Mean	Skewness	Kurtosis	Jarque-Bera
Crude oil	1803	378	−0.6382	2.1886	10.9303	1289.8890*
oil price	2958	378	−0.3232	0.8710	3.1914	47.9965*

Note: * Represents significance at the 1% level.

$$PT = \frac{P - P^*}{\sqrt{\text{Var}(p) - \text{Var}(P^*)}} \quad (10)$$

where $P^* = p_a * p_f + (1 - p_a) * (1 - p_f)$, p_a is the probability estimate of the positive change in the actual values, and p_f is the probability estimate of the positive change in the forecasting values. $P = \frac{1}{N} \sum_{t=2}^N d_t$.

$$DM = \frac{\bar{g}}{\sqrt{\hat{V}/N}} \quad (11)$$

where $\bar{g} = \sum_{t=1}^N \frac{g_t}{N}$ ($g_t = (y_t - \hat{y}_{tm})^2 - (y_t - \hat{y}_{bm})^2$) and $\hat{V} = \gamma_0 + 2 \sum_{t=1}^{\infty} \gamma_t(\gamma_t = \text{cov}(g_t - g_{t-1}))$. $\hat{y}_{tm}$ is the values forecasted by the test model, and $\hat{y}_{bm}$ represents the values forecasted by the benchmark model.

$$IR_{MAE} = \frac{(MAE_A - MAE_B)}{MAE_B} \times 100\% \quad (12)$$

$$IR_{MAPE} = \frac{(MAPE_A - MAPE_B)}{MAPE_B} \times 100\% \quad (13)$$

$$IR_{RMSE} = \frac{(RMSE_A - RMSE_B)}{RMSE_B} \times 100\% \quad (14)$$

where A and B represent two different models.

4.2. Empirical results and analysis

4.2.1. Preprocessing of finance and oil news data

(1) Preprocessing of finance data

Six financial variables (DJIA, S&P500, NASDAQ, BUSDX, USRUE, and crude oil futures closing price) are collected in the section above. However, if these variables are fed back into the multivariate decomposition method, many subcomponents will be generated, which will significantly increases the burden of testing and calculation. More importantly, there may exist multi-collinearity among these variables. For these two reasons, KPCA is employed to extract the main information of the financial variables, and Gaussian kernel function is regarded as the kernel function. The empirical results show that the cumulative variance contribution rate of the first two principal components reaches 82.60%, which indicates that the first two principal components extracted contain the most information of the financial variables. Therefore, the first two principal components are regarded as proxies for financial variables.

(2) Preprocessing of oil news text

First, tokenization and filtering out stop words and punctuations are performed for the crude oil news headlines. Further, sentiment values quantified by Python libraries (Textblob and Vader) are added and averaged. Thereupon, two daily sentiment polarity score sequences corresponding to the keywords “Crude oil” and “oil price” are achieved. Finally, the weekly sentiment polarity sequences are obtained by summing and normalizing the scores for every seven days. Table 2 displays the basic statistical descriptions for the two weekly sentiment polarity sequences, which indicates that they do not follow normal distribution.

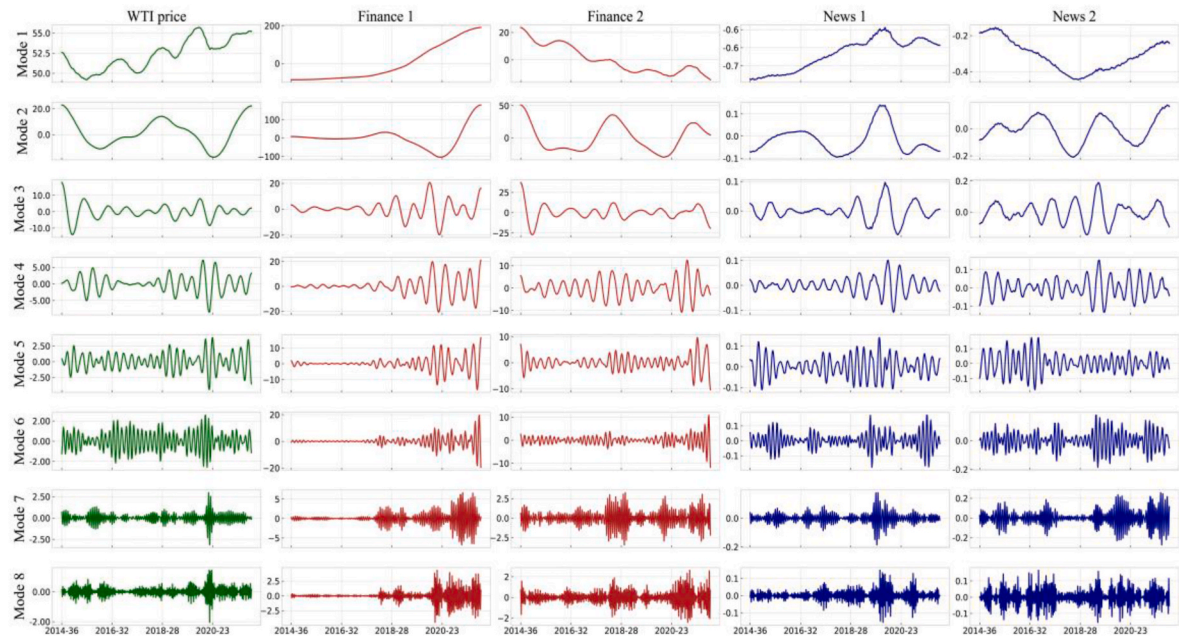


Fig. 2. First decomposition results.

Table 3
SE values for the subcomponents of WTI price after the first decomposition.

Mode	Mode 1	Mode 2	Mode 3	Mode 4
SE	0.1987	0.1709	0.6195	0.5073
Mode	Mode 5	Mode 6	Mode 7	Mode 8
SE	0.6661	0.5899	0.3707	0.4423

4.2.2. Two-layer multivariate decomposition and reconstruction

Through the steps above, two financial and news indexes are obtained, and they are denoted as “Finance 1”, “Finance 2”, “News 1”, and “News 2”. Therefore, the first decomposition is carried out for the WTI price and two financial and news indexes. The decomposition results are shown in Fig. 2. It is obvious that the frequencies of the subcomponents increase and the amplitudes of the subcomponents decrease from top to bottom.

However, too many subcomponents may increase the cumulative

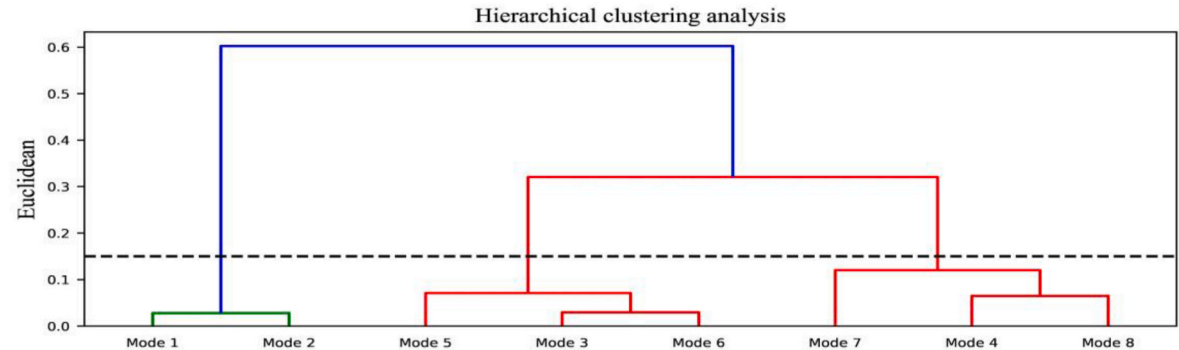


Fig. 3. HCA results for the subcomponents of WTI price after the first decomposition.

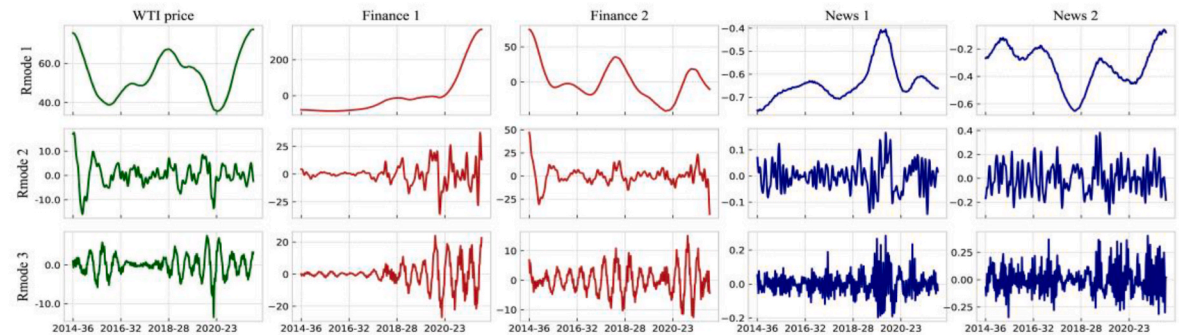


Fig. 4. First reconstruction results.

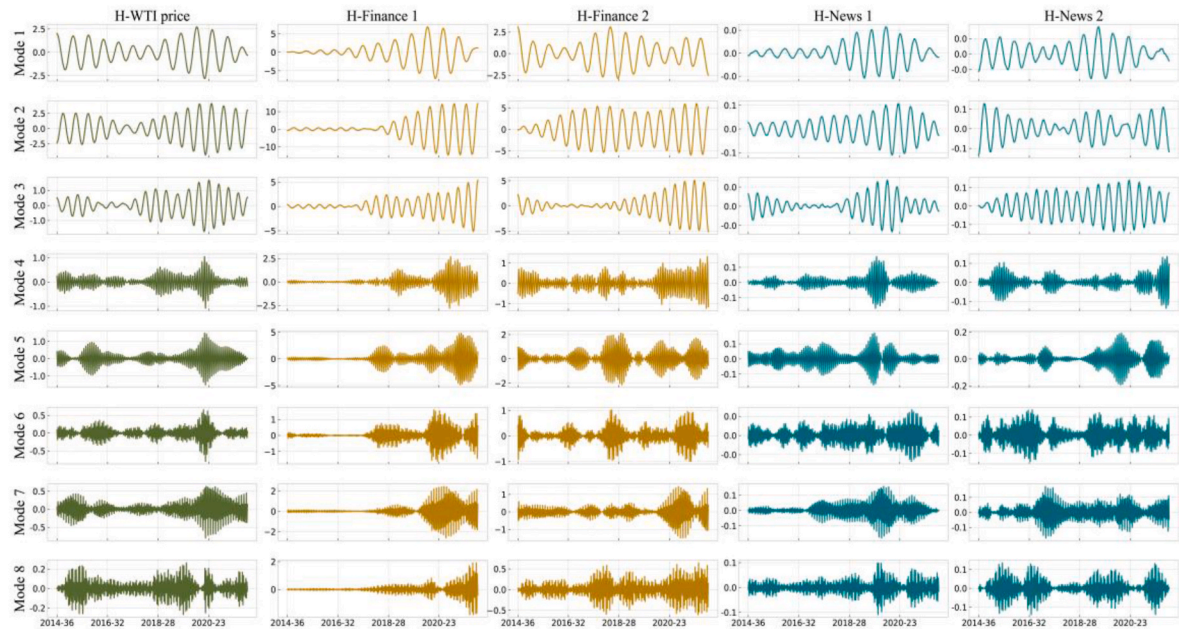


Fig. 5. Second decomposition results.

Table 4
SE values for the subcomponent of WTI price after the second decomposition.

Mode	H-mode 1	H-mode 2	H-mode 3	H-mode 4
SE	0.5936	0.6656	0.5106	0.1051
Mode	H-mode 5	H-mode 6	H-mode 7	H-mode 8
SE	0.0985	0.2046	0.3194	0.0791

error and computational burden. In light of this, the eight sub-components of crude oil prices are reconstructed according to the following two steps.

First, as the SE value represents the complexity of the sequence, the SE value is calculated to analyze complexity of each subcomponent. The results are shown in Table 3, which suggests that the complexities of modes 1 and 2, modes 3 to 6, and modes 7 to 8 are similar.

Second, from a methodological perspective, HCA is used to further validate and modify the abovementioned analysis. According to the HCA results in Fig. 3, modes 1–2 are reconstructed into low-frequency component; modes 3, 5, and 6 are reconstructed into medium-frequency component; mode 4, 7, and 8 are reconstructed into high-frequency component.

Meantime, according to the decomposition principle of MVMD, the decomposition results of the crude oil price and its corresponding forecasters are matched at each timescale. Therefore, this study

reconstructs the subcomponents of forecasters according to the reconstruction rule of the crude oil prices. Eventually, the reconstructed modes (Rmode) for all variables are shown in Fig. 4. Compared with Fig. 2, the frequency characteristics in Fig. 4 are more obvious.

The second decomposition and reconstruction procedure are performed for high-frequency subcomponents, where the high-frequency subcomponents are denoted as “H-WTI price”, “H-Finance 1”, “H-Finance 2”, “H-News 1”, and “H-News 2”. First, Fig. 5 shows the results of the second MVMD decomposition. Second, the SE values of all the subcomponents are calculated, as shown in Table 4. The results suggest that the modes 1–3 can be reconstructed to low-frequency, modes 4–5 and mode 8 can be reconstructed to medium-frequency, and modes 6–7 can be reconstructed to high-frequency. These reconstruction rules are also verified by HCA, and as shown in Fig. 6. Finally, the reconstruction results are displayed in Fig. 7. It should be noted that, to show the frequency characteristics more clearly, the sub-modes of the high-frequency subcomponent of crude oil price are magnified.

4.2.3. Feature selection

(1) Screening effective forecasters

Effective forecasters are screened using statistical methods. Specifically, the stationarity of all subcomponents is first tested by Augmented

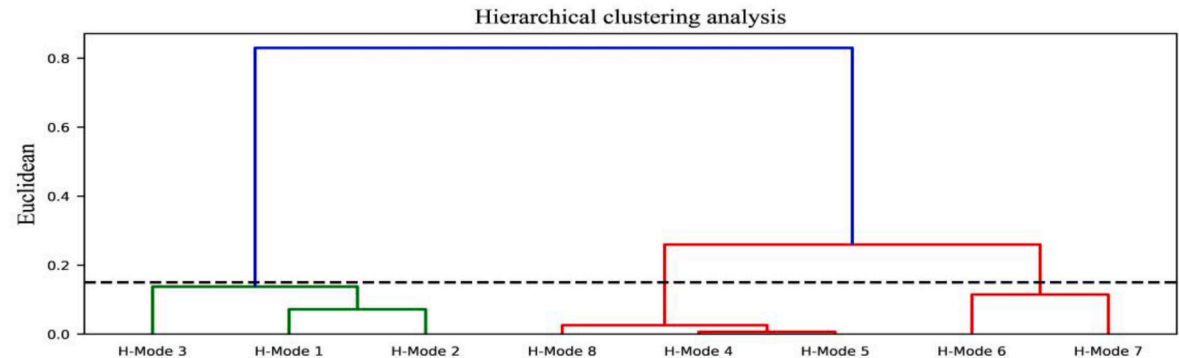


Fig. 6. HCA results for the subcomponents of WTI price after the second decomposition.

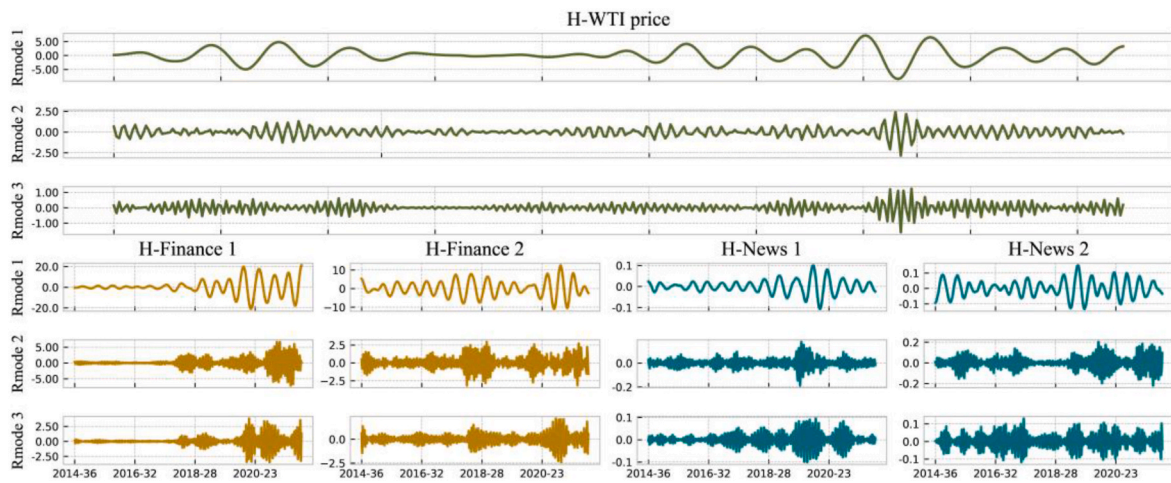


Fig. 7. Second reconstruction results.

Table 5

Statistical test results for the subcomponents after the first decomposition and reconstruction.

Panel A: ADF test at the 5% significance level					
Variables	At the original level	At the first-order difference	Variables	At the original level	At the first-order difference
WTI-Rmode 1	-2.9280 (0.0421)		News 2-Rmode 3	-8.4890 (0.0000)	
WTI-Rmode 2	-5.8124 (0.0000)		Finance 1-Rmode 1	-2.4386 (0.3593)	-3.0646 (0.1148)
WTI-Rmode 3	-8.0078 (0.0000)		Finance 1-Rmode 2	-6.5055 (0.0000)	
News 1-Rmode 1	-1.8492 (0.3563)	-4.1883 (0.0006)	Finance 1-Rmode 3	-7.1375 (0.0000)	
News 1-Rmode 2	-5.3887 (0.0000)		Finance 2-Rmode 1	-2.1172 (0.2376)	-3.8574 (0.0023)
News 1-Rmode 3	-8.7605 (0.0000)		Finance 2-Rmode 2	-5.6237 (0.0000)	
News 2-Rmode 1	-2.7346 (0.0682)	-3.2735 (0.0161)	Finance 2-Rmode 3	-8.1849 (0.0000)	
News 2-Rmode 2	-6.5909 (0.0000)				
Panel B: Granger causality test at the 5% significance level					
		lag (-1)	lag (-2)	lag (-3)	lag (-4)
WTI-Rmode 2	News 1-Rmode 2	0.6910 (0.4064)	2.2493 (0.1069)	4.0839 (0.0072)	4.7540 (0.0009)
	News 2-Rmode 2	26.9579 (0.0000)	13.7027 (0.0000)	6.4734 (0.0003)	5.4365 (0.0003)
	Finance 1-Rmode 2	0.2032 (0.6524)	0.7515 (0.4724)	1.3692 (0.2519)	1.2027 (0.3092)
	Finance 2-Rmode 2	1.3965 (0.2381)	2.5828 (0.0769)	8.0036 (0.0000)	2.0385 (0.0885)
WTI-Rmode 3	News 1-Rmode 3	0.6259 (0.4293)	0.1683 (0.8452)	4.4195 (0.0045)	3.9942 (0.0035)
	News 2-Rmode 3	7.5210 (0.0064)	3.3344 (0.0367)	3.1221 (0.0260)	4.4244 (0.0017)
	Finance 1-Rmode 3	0.8304 (0.3628)	0.6368 (0.5295)	0.4397 (0.7248)	1.1598 (0.3283)
	Finance 2-Rmode 3	2.6381 (0.1052)	1.9201 (0.1480)	1.4788 (0.2199)	4.3581 (0.0019)

Note: The values in parentheses in the twelfth line represent the lag order of variables.

Table 6

Statistical test results for the subcomponents after the second decomposition and reconstruction.

Panel A: ADF test at the 5% significance level					
WTI-HRmode 1	News 1-HRmode 1	News 2-HRmode 1	Finance 1-HRmode 1	Finance 2-HRmode 1	
-8.0645 (0.0000)	-11.9901 (0.0000)	-11.3371 (0.0000)	-9.8697 (0.0000)	-7.2150 (0.0000)	
WTI-HRmode 2	News 1-HRmode 2	News 2-HRmode 2	Finance 1-HRmode 2	Finance 2-HRmode 2	
-4.7560 (0.0000)	-13.1329 (0.0000)	-14.7633 (0.0000)	-4.7009 (0.0000)	-3.9680 (0.0001)	
WTI-HRmode 3	News 1-HRmode 3	News 2-HRmode 3	Finance 1-HRmode 3	Finance 2-HRmode 3	
-3.6422 (0.0002)	-5.4854 (0.0000)	-16.1390 (0.0000)	-3.3528 (0.0008)	-3.2611 (0.0011)	
Panel B: Granger causality test at the 5% significance level					
		lag (-1)	lag (-2)	lag (-3)	
WTI-HRmode 1	News 1-HRmode 1	66.1621 (0.0000)	7.6941 (0.0005)	3.3227 (0.0199)	
	News 2-HRmode 1	7.3722 (0.0069)	40.0002 (0.0000)	16.1272 (0.0000)	
	Finance 1-HRmode 1	0.2960 (0.5867)	4.1380 (0.0167)	2.9995 (0.0306)	
	Finance 2-HRmode 1	11.2409 (0.0009)	12.2712 (0.0000)	4.6337 (0.0034)	
WTI-HRmode 2	News 1-HRmode 2	0.2215 (0.6381)	0.5342 (0.5866)	3.9364 (0.0087)	
	News 2-HRmode 2	45.5294 (0.0000)	1.9856 (0.1388)	2.9468 (0.0328)	
	Finance 1-HRmode 2	0.2088 (0.6480)	0.6858 (0.5043)	2.4003 (0.0675)	
	Finance 2-HRmode 2	16.3676 (0.0001)	0.8733 (0.4184)	0.2574 (0.8560)	
WTI-HRmode 3	News 1-HRmode 3	0.0000 (0.9964)	0.6303 (0.5330)	3.1427 (0.0253)	
	News 2-HRmode 3	4.1839 (0.0415)	0.2692 (0.7641)	1.6121 (0.1861)	
	Finance 1-HRmode 3	1.7242 (0.1900)	17.0900 (0.0000)	8.7528 (0.0000)	
	Finance 2-HRmode 3	23.5427 (0.0000)	11.5436 (0.0000)	14.1089 (0.0000)	

Note: Panel A is the test results of all the original series (high frequency reconstruction subcomponents).

Dickey-Fuller (ADF) test. Then co-integration and Granger causality test are conducted according to the stationary characteristic of the tested series.

The statistical test results for the subcomponents after the first decomposition and reconstruction are presented in Table 5. According

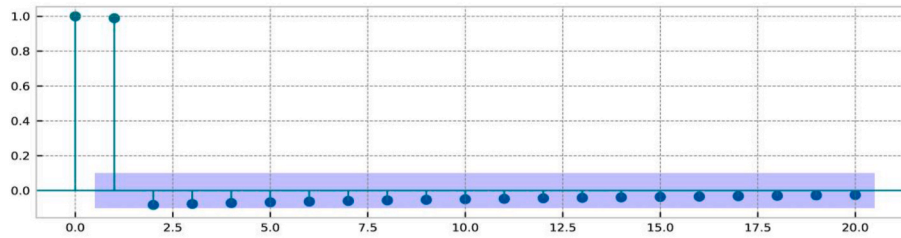


Fig. 8. PACF result for the WTI-Rmodel 1.

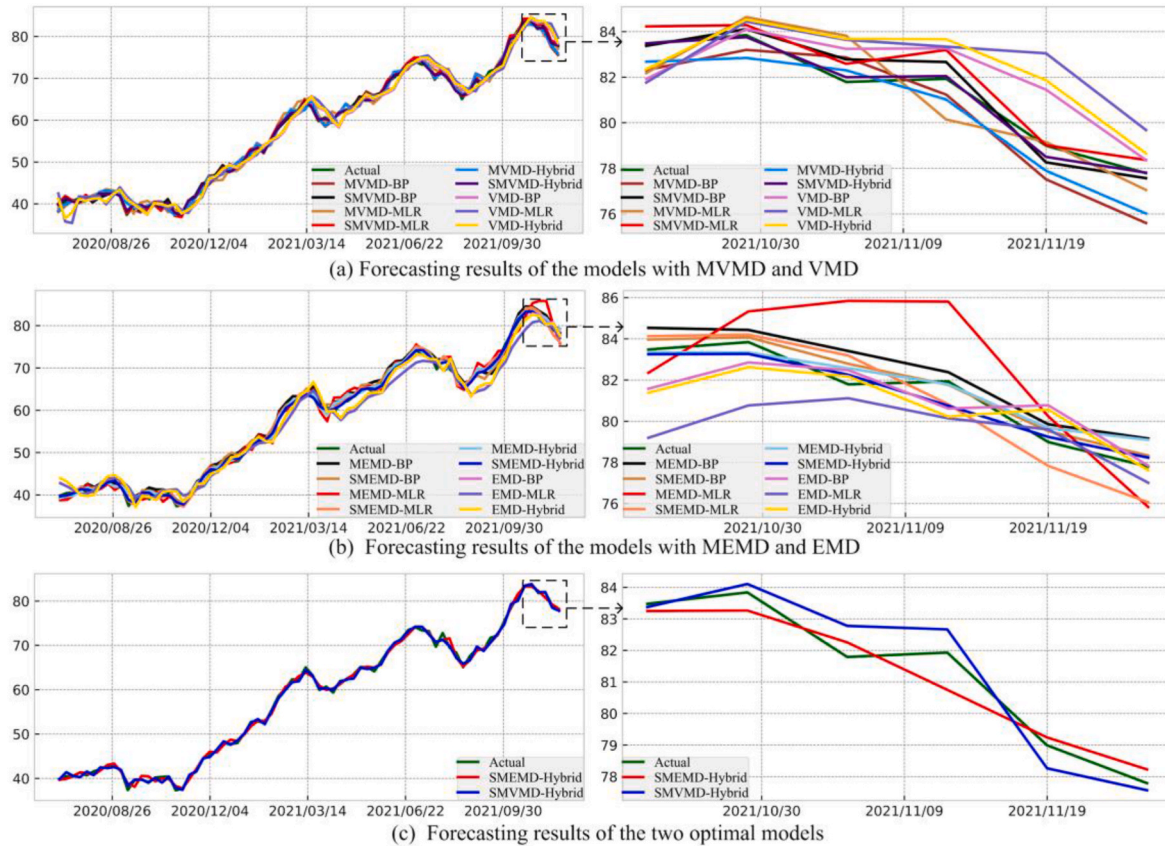


Fig. 9. Forecasting results for all the models.

(2) Forecasting performance evaluation

(a) Forecasting accuracy

to the *Panel A*, News 1-Rmode 1, News 2-Rmode 1, and Finance 2-Rmode 1 are first order difference stationary, Finance 1-Rmode 1 are not first order difference stationary, and other variables are original series stationary. That is, WTI-Rmode 1 and its forecasters are not satisfied with the prerequisite of the co-integration test. Therefore, the Granger causality test is only implemented for variables with the original series stationary, and the test results are shown in *Panel B*. The results indicate that News 1-Rmode 2, News 2-Rmode 2, and Finance 2-Rmode 2 can be used as effective forecasters of WTI-Rmode 2; News 1-Rmode 3, News 2-Rmode 3, and Finance 2-Rmode 3 can be used as effective forecasters of WTI-Rmode 3.

The statistical test results for the subcomponents after the second decomposition and reconstruction are shown in *Table 6*. From the *Panel A*, it can be seen that all variables are original series stationary, which means that the Granger causality test can be performed for all the subcomponents of crude oil prices and their corresponding forecasters. The results in *Panel B* indicate that, except for WTI-HRmode 2 and Finance 1-HRmode 2, there are Granger causalities between the predictive

subcomponents and their corresponding forecasters. In other words, Finance 1-HRmode 2 can not be used as an effective forecaster of WTI-HRmode 2, and other forecasters can be used as effective forecasters of the corresponding predictive targets.

(2) Determining optimal lag order

This study adopts the lag criterion of VAR model to determine the optimal lag order for the subcomponents of crude oil prices with effective forecasters. Eventually, the optimal lag order of the subcomponents after the first decomposition and reconstruction is four, and the optimal lag order of the subcomponents after the second decomposition and reconstruction is three. For the subcomponents of crude oil prices without effective forecasters, the chaotic characteristics are first judged. Among all the subcomponents of crude oil price, only WTI-Rmode 1 has no corresponding effective forecasters, and its MLYE is -0.0042 . Therefore, its optimal lag order is determined to be the four using PACF, as shown in *Fig. 8*.

Table 7

Forecasting accuracy evaluation results for all models.

Model	MAE	MAPE	RMSE	DA	PT test
MEMD-BPNN	1.1197	2.0352	1.3513	0.7432	3.2735 (0.0005)
MEMD-MLR	1.2217	2.1231	1.5484	0.7567	2.3611 (0.0091)
MEMD-Hybrid	0.7578	1.3832	0.9585	0.7432	2.7389 (0.0030)
MVMD-BPNN	1.0070	1.8053	1.1983	0.7703	3.6068 (0.0001)
MVMD-MLR	1.0844	1.9704	1.3674	0.7837	2.0175 (0.0218)
MVMD-Hybrid	0.8990	1.6009	1.0971	0.7837	3.6068 (0.0001)
EMD-BPNN	1.6463	2.9463	2.0296	0.6621	3.0770 (0.0010)
EMD-MLR	1.8981	3.1998	2.3570	0.6041	2.4566 (0.0070)
EMD-Hybrid	1.6383	2.9354	2.0195	0.6756	3.0770 (0.0010)
SMEMD-BPNN	0.8067	1.5063	0.9820	0.8108	5.1961 (0.0000)
SMEMD-MLR	0.6832	1.2446	0.8586	0.8108	4.9886 (0.0000)
SMEMD-Hybrid	0.5955	1.0920	0.7230	0.8108	4.0170 (0.0000)
VMD-BPNN	1.3877	2.5212	1.7000	0.6756	2.7389 (0.0030)
VMD-MLR	1.6451	3.0221	2.0450	0.5810	2.1798 (0.0146)
VMD-Hybrid	1.3532	2.4636	1.6576	0.6891	2.5472 (0.0054)
SMVMD-BPNN	0.6104	1.0958	0.7468	0.8378	5.1961 (0.0000)
SMVMD-MLR	0.5725	1.0160	0.7293	0.8783	5.7259 (0.0000)
SMVMD-Hybrid	0.4618	0.8236	0.5866	0.8919	6.0396 (0.0000)

4.2.4. Crude oil price forecasting and performance evaluation

(1) Crude oil price forecasting

The final forecasting results are shown in Fig. 9, in which the three subgraphs on the left are the forecasting values of all the test samples, and three subgraphs on the right are forecasting values for the last two months of the test sample. The dark green and purple trend lines are the most consistent in subgraph (a), and the dark green and blue trend lines are the most consistent in subgraph (b). In other words, the hybrid forecasting model with the multivariate decomposition method has the best forecasting ability. From subgraph (c), it can be seen that the SMVMD-Hybrid (S represents second) model outperforms the SMEMD-Hybrid model, namely the proposed model (SMVMD-Hybrid) has the best forecasting performance.

Table 7 shows the forecasting accuracy of all the models, which indicates that the level and direction forecasting accuracy of SMVMD-Hybrid is the best. Meantime, it can be seen that the forecasting accuracy of the models with the secondary decomposition is better than that of the models with first decomposition. For example, in lines 4 and 13, the MAE value decreases from 0.7578 to 0.5955. This finding extends the idea of the first decomposition-ensemble research [32]. The forecasting accuracy of the model with multivariate decomposition is better than that of the model with univariate decomposition. For example, the MAE, MAPE, and RMSE of the VMD-BPNN are 1.3877, 2.5212, and 1.7000, respectively, and that of MVMD-BPNN are 1.0070, 1.8053, and

1.1983, respectively; thus the forecasting accuracy is significantly improved. This result also validates the conclusion obtained in the literature [27]. It is also found that although the forecasting accuracy of MEMD-Hybrid is better than that of MVMD-Hybrid, overall, the forecasting accuracy of the model with MVMD is better than that of the model with MEMD. For direction forecasting performance, the models with the second multivariate decomposition have the highest direction forecasting accuracy, followed by the models with the first multivariate decomposition, and finally, the models with the univariate decomposition.

(b) Significance and effectiveness

First, DM test is performed to verify the significance of the proposed model, and MAE, RMSE, and MAPE are selected as its loss functions. Meantime, the SMVMD-Hybrid is regarded as the test models, and the other models are used as the benchmark models. Table 8 displays the DM test results, which indicate that the SMVMD-Hybrid model is superior to the SMVMD-MLR and SMEMD-Hybrid models at the significance level of 5% and 10%; the p-values of the other models are less than 0.01, that is, the model proposed by this study is optimal.

Second, IR is calculated to evaluate the effectiveness of the proposed model. Table 9 shows the IR results between the proposed model and the benchmark models. It can be seen that all the IR values are greater than 10%, which indicate that the proposed model is superior to the other benchmark models. Moreover, as SMEMD-Hybrid, SMVMD-BPNN, and SMVMD-MLR have a relatively small IR, their forecasting ability are second only to the proposed model.

The abovementioned part analyzes the significance and validity of the proposed model. In the following section, we analyze the impacts of different decomposition times, decomposition methods, and forecasting

Table 9

IR results for different benchmark models.

Benchmark model → Proposed model	IR _{MAE}	IR _{MAPE}	IR _{RMSE}
MEMD-BPNN → SMVMD-Hybrid	−65.79%	−121.16%	−76.47%
MEMD-MLR → SMVMD-Hybrid	−75.99%	−129.95%	−96.18%
MEMD-Hybrid → SMVMD-Hybrid	−29.60%	−55.96%	−37.19%
MVMD-BPNN → SMVMD-Hybrid	−55.30%	−101.16%	−62.28%
MVMD-MLR → SMVMD-Hybrid	−62.26%	−114.68%	−78.08%
MVMD-Hybrid → SMVMD-Hybrid	−43.72%	−77.73%	−51.05%
SMEMD-BPNN → SMVMD-Hybrid	−34.49%	−68.27%	−39.54%
SMEMD-MLR → SMVMD-Hybrid	−22.14%	−42.10%	−27.20%
SMEMD-Hybrid → SMVMD-Hybrid	−13.37%	−26.84%	−13.64%
SMVMD-BPNN → SMVMD-Hybrid	−14.86%	−27.22%	−16.02%
SMVMD-MLR → SMVMD-Hybrid	−11.07%	−19.24%	−14.27%

Table 8

DM test results for different benchmark models.

Benchmark model	DM _{MAE}	P _{MAE} -value	DM _{MSE}	P _{MSE} -value	DM _{MAPE}	P _{MAPE} -value
MEMD-BPNN	−9.8983	0.0000*	−8.3006	0.0000*	−7.8412	0.0000*
MEMD-MLR	−5.4549	0.0000*	−4.0141	0.0001*	−6.3653	0.0000*
MEMD-Hybrid	−4.5338	0.0000*	−3.9454	0.0001*	−4.2083	0.0000*
MVMD-BPNN	−6.2461	0.0000*	−5.7005	0.0000*	−5.6489	0.0000*
MVMD-MLR	−6.1823	0.0000*	−5.1546	0.0000*	−5.1757	0.0000*
MVMD-Hybrid	−5.2107	0.0000*	−5.0655	0.0000*	−5.2408	0.0000*
EMD-BPNN	−7.1761	0.0000*	−5.0187	0.0000*	−6.9161	0.0000*
EMD-MLR	−6.4431	0.0000*	−5.0011	0.0000*	−6.8512	0.0000*
EMD-Hybrid	−7.2166	0.0000*	−5.1317	0.0000*	−6.8811	0.0000*
SMEMD-BPNN	−4.6083	0.0000*	−4.1634	0.0000*	−4.0268	0.0001*
SMEMD-MLR	−2.9099	0.0047*	−2.6989	0.0086*	−2.9443	0.0043*
SMEMD-Hybrid	−2.0521	0.0437**	−1.8588	0.0670***	−1.9455	0.0555***
VMD-BPNN	−7.5298	0.0000*	−6.4446	0.0000*	−6.2289	0.0000*
VMD-MLR	−7.5315	0.0000*	−5.1975	0.0000*	−6.0683	0.0000*
VMD-Hybrid	−7.0273	0.0000*	−6.0733	0.0000*	−5.9305	0.0000*
SMVMD-BPNN	−3.8844	0.0002*	−3.2793	0.0015*	−3.6365	0.0005*
SMVMD-MLR	−2.5933	0.0114**	−3.1074	0.0026*	−2.5486	0.0129**

Notes: *, **, and *** mean significance at the 1%, 5%, and 10% levels, respectively.

Table 10
DM test results with different test models.

Benchmark model	Test model	DM test	P-value
Panel A: The significance comparison of the models with different decomposition times			
MVMD-BPNN	SMVMD-BPNN	-4.1560	0.0000*
MVMD-MLR	SMVMD-MLR	-3.9356	0.0001*
MVMD-Hybrid	SMVMD-Hybrid	-5.0655	0.0000*
MEMD-BPNN	SMEEMD-BPNN	-5.4719	0.0000*
MEMD-MLR	SMEEMD-MLR	-3.5151	0.0007*
MEMD-Hybrid	SMEEMD-Hybrid	-3.6736	0.0004*
Panel B: The significance comparison of the models with different decomposition methods			
MEMD-BPNN	MVMD-BPNN	-2.2333	0.0285 **
MEMD-MLR	MVMD-MLR	-1.2463	0.2166
MEMD-Hybrid	MVMD-Hybrid	2.2442	-0.0278
SMEEMD-BPNN	SMVMD-BPNN	-2.6333	0.0103**
SMEEMD-MLR	SMVMD-MLR	-1.2961	0.1990
SMEEMD-Hybrid	SMVMD-Hybrid	-1.8588	0.0670***
EMD-BPNN	MEMD-BPNN	-3.1773	0.0021*
EMD-MLR	MEMD-MLR	-2.7022	0.0085*
EMD-Hybrid	MEMD-Hybrid	-4.5528	0.0000*
VMD-BPNN	MVMD-BPNN	-3.2783	0.0016*
VMD-MLR	MVMD-MLR	-3.2243	0.0018*
VMD-Hybrid	MVMD-Hybrid	-3.7073	0.0004*
Panel C: The significance comparison of the models with different forecasting techniques			
MVMD-BPNN	MVMD-Hybrid	-2.4689	0.0158**
MVMD-MLR	MVMD-Hybrid	-2.4400	0.0171**
SMVMD-BPNN	SMVMD-Hybrid	-3.2793	0.0015*
SMVMD-MLR	SMVMD-Hybrid	-3.1074	0.0026*
MEMD-BPNN	MEMD-Hybrid	-7.3648	0.0000*
MEMD-MLR	MEMD-Hybrid	-3.1063	0.0026*
SMEEMD-BPNN	SMEEMD-Hybrid	-3.3673	0.0012*
SMEEMD-MLR	SMEEMD-Hybrid	-2.2227	0.0293**

Notes:*, **, and *** mean significance at the 1%, 5%, and 10% levels, respectively.

techniques on the forecasting performance of crude oil prices.

First, DM test is performed, and the results are shown in Table 10. Panel A is the DM test results with different decomposition times. The results show that the forecasting performance of the models with secondary decomposition are significantly better than that of the models with first decomposition. Panel B shows the test results with different decomposition methods. It can be found that when MVMD-MLR, SMVMD-Hybrid, and SMVMD-MLR are used as test models, they do not pass DM test. Meantime, only the MEMD-Hybrid passes the DM test when we conduct the reverse test, which indicates that the model with MVMD still has some advantages over the model with MEMD, and this advantage is not obvious. When the models with the univariate decomposition method are used as the benchmark models, all p-values are less than 0.01, namely the models with the multivariate decomposition significantly outperform the models with the univariate decomposition. Panel C shows the test results with different forecasting techniques, which suggest that the forecasting performance of the models with the hybrid forecasting technique are significantly better than that of the models with the single forecasting technique.

Second, IR value is calculated. As can be seen from the results (see Table 11), the average IR of the models with multivariate decomposition in Panel B is 68.01%, 119.92%, and 83.55%, respectively, which means that the multivariate decomposition method contributes the most to the forecasting accuracy of crude oil prices. Following the models with the second decomposition in Panel A, the average IR values are 39.46%, 69.50%, and 48.43%. Next, the models with hybrid forecasting technique in Panel C, the average IR values are 21.07%, 37.84%, and 25.79%, respectively. Finally, the model with MVMD in Panel B, the average IR is 9.03%, 17.38%, and 11.42%, respectively.

Table 11
Improvement percentage for different models.

Benchmark model	IRMAE	IRMAPE	IRRMSE
Panel A: The effectiveness comparison of the models with different decomposition times			
MVMD-BPNN → SMVMD-BPNN	-40.44%	-73.94%	-46.26%
MVMD-MLR → SMVMD-MLR	-51.19%	-95.44%	-63.81%
MVMD-Hybrid → SMVMD-Hybrid	-43.72%	-77.73%	-51.05%
MEMD-BPNN → SMEEMD-BPNN	-31.30%	-52.89%	-36.93%
MEMD-MLR → SMEEMD-MLR	-53.85%	-87.85%	-68.98%
MEMD-Hybrid → SMEEMD-Hybrid	-16.23%	-29.12%	-23.55%
Average IR	-39.46%	-69.50%	-48.43%
Panel B: The effectiveness comparison of the models with different decomposition methods			
MEMD-BPNN → MVMD-BPNN	-10.49%	-20.00%	-14.19%
MEMD-MLR → MVMD-MLR	-13.73%	-15.27%	-18.10%
MEMD-Hybrid → MVMD-Hybrid	14.12%	21.77%	13.86%
SMEEMD-BPNN → SMVMD-BPNN	-19.63%	-41.05%	-23.52%
SMEEMD-MLR → SMVMD-MLR	-11.07%	-22.86%	-12.93%
SMEEMD-Hybrid → SMVMD-Hybrid	-13.37%	-26.84%	-13.64%
Average IR	-9.03%	-17.38%	-11.42%
EMD-BPNN → MEMD-BPNN	-52.66%	-91.11%	-67.83%
EMD-MLR → MEMD-MLR	-67.64%	-107.67%	-80.86%
EMD-Hybrid → MEMD-Hybrid	-88.05%	-155.22%	-106.10%
VMD-BPNN → MVMD-BPNN	-58.10%	-101.49%	-71.80%
VMD-MLR → MVMD-MLR	-96.19%	-177.75%	-118.64%
VMD-Hybrid → MVMD-Hybrid	-45.42%	-86.27%	-56.05%
Average IR	-68.01%	-119.92%	-83.55%
Panel C: The effectiveness comparison of the models with different forecasting techniques			
MVMD-BPNN → MVMD-Hybrid	-11.58%	-23.43%	-11.23%
MVMD-MLR → MVMD-Hybrid	-18.54%	-36.95%	-27.03%
SMVMD-BPNN → SMVMD-Hybrid	-14.86%	-27.22%	-16.02%
SMVMD-MLR → SMVMD-Hybrid	-11.07%	-19.24%	-14.27%
MEMD-BPNN → MEMD-Hybrid	-36.19%	-65.20%	-39.28%
MEMD-MLR → MEMD-Hybrid	-46.39%	-73.99%	-58.99%
SMEEMD-BPNN → SMEEMD-Hybrid	-21.12%	-41.43%	-25.90%
SMEEMD-MLR → SMEEMD-Hybrid	-8.77%	-15.26%	-13.56%
Average IR	-21.07%	-37.84%	-25.79%

5. Conclusion and further study

This study proposes a novel hybrid forecasting model with two-layer multivariate decomposition. For practical purposes, the weekly WTI spot price is employed to verify the forecasting performance of the proposed model, and some meaningful conclusions are obtained. (1) The models with the second multivariate decomposition are obviously better than the models with the first decomposition. The main reason for this result is that the high-frequency components, which are difficult to forecast, are also decomposed into several simpler modes. (2) Compared with the univariate decomposition method, the multivariate decomposition method can significantly improve forecasting accuracy of crude oil prices. The results are mainly attributed to the fact that multivariate decomposition methods simultaneously decompose multiple complex series into several simple series. (3) The forecasting performance of the models with MVMD is not significantly better than that of the models with MEMD. The main reason may be related to the number of MVMD subcomponents. (4) The forecasting effect of the models with the hybrid forecasting technique is obviously better than that of the models with the single forecasting technique, mainly because different forecasting techniques are used for the subcomponents with different characteristics. (5) The proposed model (SMVMD-Hybrid) has the best forecasting performance.

For the implications of this study to energy policy and markets, our study can provide some inspiration. First, the model proposed has good forecasting performance. Hence, it can provide a new research paradigm for the forecasting of other energy market (such as natural gas and electricity). Meantime, as the data used in the proposed model can be freely acquired from publicly available websites, crude oil market

participants can collect relevant data and adopt our proposed forecasting framework to forecasting trend change of crude oil prices. Second, accurate crude oil price forecasting can provide meaningful guidance for the government to make reasonable exchange rate and energy market regulation policy. Third, accurate crude oil price forecasting can provide scientific reference for the pricing and decision-making of crude oil producers, and can also help other crude oil market participants to develop reasonable trading strategies.

Nevertheless, this study also has some limitations and potential extensions. First, this study uses only one case to verify the proposed model. In the future, it will be necessary to verify the proposed model in other similar markets, such as carbon, electricity, and natural gas prices. Second, the parameters of the most methods used in this study are determined through trial and error, which usually takes a lot of time and effort. In view of this, we can try to choose some good intelligent algorithms to optimize and determine them in the later stage. Third, how to use multiscale decomposition method to decompose samples is a difficult and valuable research issue. Finally, the interval forecasting of crude oil prices is an important research work in the future.

CRedit authorship contribution statement

Zhengling Zhao: Conceptualization, Software, Formal analysis, Writing- original draft. **Jingyun Sun:** Writing - review & editing, Funding acquisition. **Shaolong Sun:** Methodology, Data Curation. **Wang Shouyang:** Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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